Input-Token Cost of Context Compaction: Definitions, Assumptions, and the Optimal Compaction Threshold

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1 Definitions

Definition 1 (Prices and input cost)**.** Cache-read, cache-write, and output tokens are priced at u, αu, and βu. At turn $t = 1, \ldots, T$, let P_t be the prompt length, and R_t, W_t the cache-read and cache-write tokens, with $P_t = R_t + W_t$. Write $S_{cr} = \sum_t R_t$, $S_{cw} = \sum_t W_t$, and $A = \sum_t P_t$. The input cost is

$$\text{Cost}_{\text{in}} = u\,(S_{cr} + \alpha S_{cw}).$$

Definition 2 (Compaction policy and threshold)**.** When the prompt about to be sent exceeds the threshold, the context is compacted in that turn. L denotes the prompt length just before compaction (the peak) and is the decision variable.

Definition 3 (Compacted context)**.** After compaction the prompt is $L_0 + L_p$, with $L_p = S + Q$, where L_0 is the unchanged prefix (system prompt and task), S is the summary of the earlier history, and Q is the recent history kept verbatim.

Definition 4 (Window)**.** A window is the sequence of turns from one compaction turn up to, but not including, the next one. K denotes the number of turns in a window.

Definition 5 (Per-turn cost)**.** $f(L)$ is the average input cost per turn, in units of u, when the threshold is L.

2 Assumptions

Assumption 1 (Cache)**.** The prefix cache never expires or is evicted. On a turn without compaction the whole previous request stays cached; on a compaction turn only L_0 stays cached.

Assumption 2 (Growth per turn)**.** On every turn without compaction the prompt grows by g, and the write is taken to be g as well ($W_t \approx g$).

Assumption 3 (Compaction turn)**.** The compaction turn also advances the task by one turn, and the new content of that turn is contained in Q. Hence the write on a compaction turn is L_p.

Assumption 4 (Shape of L_p)**.** The summary and the verbatim part are each either fixed or proportional to the peak L, and do not depend on how many compactions have already occurred:

$$S = s_0 + s_1 L, \qquad Q = q_0 + q_1 L \implies L_p = c + \varphi L, \quad c = s_0 + q_0, \quad \varphi = s_1 + q_1, \quad 0 \le \varphi < 1.$$

A capped summary with a fixed token budget for recent turns gives $\varphi = 0$; keeping a fraction ρ of the threshold verbatim gives $q_1 = \rho$; rebuilding the summary from the full original history gives $s_1 > 0$.

Assumption 5 (Trajectory invariance)**.** The number of turns T and the new content of each turn do not depend on L.

Assumption 6 (Steady state)**.** Windows repeat many times, so differences in the first and last windows are neglected.

3 Derivation

1. Cost in terms of total prompt and writes

Since $S_{cr} = A - S_{cw}$,

$$\frac{\text{Cost}_{\text{in}}}{u} = A + (\alpha - 1)\, S_{cw}. \tag{1}$$

Every prompt token is counted once at the read price, and written tokens carry an extra $\alpha - 1$.

2. One window

By Assumptions 1–3, a window starts at the compaction turn with prompt $L_0 + L_p$ and grows by g per turn until it reaches L:

$$K = \frac{L - L_0 - L_p}{g} + 1, \qquad A_w = \frac{K\,(L_0 + L_p + L)}{2}, \qquad W_w = L_p + (K-1)\, g. \tag{2}$$

3. Per-turn cost

By Assumptions 5 and 6, the number of windows is $N = T/K$, so

$$\frac{\text{Cost}_{\text{in}}}{u} \approx N\left[A_w + (\alpha-1)W_w\right] = T\, f(L), \qquad f(L) = \frac{L_0 + L_p + L}{2} + (\alpha - 1)\left[g + \frac{L_p - g}{K}\right]. \tag{3}$$

The first term is the average prompt length and the second is the write premium on the average write per turn. Because T is a constant factor independent of L, minimizing the total cost and minimizing $f(L)$ give the same L.

4. Optimum for constant L_p ($\varphi = 0$)

Let $y = Kg = L - L_0 - L_p + g$ be the progress per window. Then

$$f = \underbrace{L_0 + L_p - \tfrac{g}{2} + (\alpha - 1)g}_{\text{independent of } L} + \underbrace{\frac{y}{2}}_{\text{re-reading}} + \underbrace{\frac{(\alpha-1)\, g\,(L_p - g)}{y}}_{\text{extra write per compaction}}. \tag{4}$$

For $L_p > g$, setting $df/dy = 0$ gives

$$y^* = \sqrt{2(\alpha-1)\, g\,(L_p - g)}, \qquad \boxed{L^* = L_0 + L_p - g + \sqrt{2(\alpha-1)\, g\,(L_p - g)}}. \tag{5}$$

At the optimum the two y-dependent terms are equal, and a deviation costs, per turn,

$$f(y) - f(y^*) = \frac{(y - y^*)^2}{2y}. \tag{6}$$

This is the structure of the economic order quantity (EOQ): the order cost is $(\alpha - 1)(L_p - g)$, the order quantity is y, and the holding cost is one read per token per turn.

5. Optimum for $L_p = c + \varphi L$

Let $D = Kg = (1-\varphi)L - L_0 - c + g$. Then

$$f(L) = \frac{L_0 + c + (1+\varphi)L}{2} + (\alpha - 1)g + \frac{(\alpha-1)\,g\,(c + \varphi L - g)}{D}, \tag{7}$$

$$\frac{d}{dL}\,\frac{c + \varphi L - g}{D} = -\frac{X}{D^2}, \qquad X = c - g + \varphi L_0. \tag{8}$$

If $X > 0$, then $f'(L) = 0$ gives $D^{*2} = 2(\alpha-1)gX/(1+\varphi)$ and

$$\boxed{L^* = \frac{L_0 + c - g + \sqrt{2(\alpha-1)\,g\,X/(1+\varphi)}}{1-\varphi}}. \tag{9}$$

For $\varphi = 0$ this reduces to (5). If $X \le 0$ there is no interior optimum.

6. Relaxing the assumptions

(a) Short tasks (relaxing Assumption 6). With current prompt C and R turns remaining, compact (approximately) if and only if

$$R\,(C - L_0 - L_p) > (\alpha - 1)(L_p - g). \tag{10}$$

The left side is the reading saved over the remaining turns; the right side is the extra write premium of one compaction.

(b) Turn count depending on L (relaxing Assumption 5). If the number of turns is $T(L)$, the total cost is $T(L)\,f(L)$ and the optimum satisfies

$$\frac{f'(L)}{f(L)} = -\frac{T'(L)}{T(L)}. \tag{11}$$

If T scales by the same factor for every L, L^* is unchanged; if smaller L produces more turns ($T' < 0$), L^* moves to larger values.

Notes

Remark (g versus w). The actual write on a normal turn, w, differs slightly from the growth g, which is the previous output plus the new observation. The previous output is cached as it is generated, which makes w smaller; the cache stores only full blocks, so the partially filled last block is rewritten, which makes w larger. Keeping w separate replaces $(L_p - g)$ in (5) by $(L_p - w)$ and X in (9) by $c - w + \varphi(L_0 + w - g)$. In our measurements the difference was small, so we set $w = g$.

Remark (Trajectory invariance). Compaction can change the number of turns and the content of later turns. We first assume invariance and later correct for it by measuring $T(L)/T_0$ at a few candidate thresholds and applying (11).